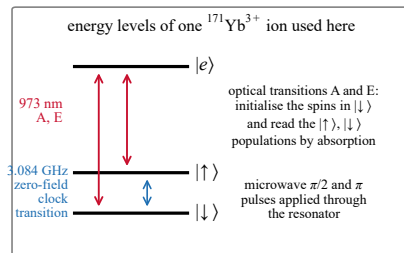
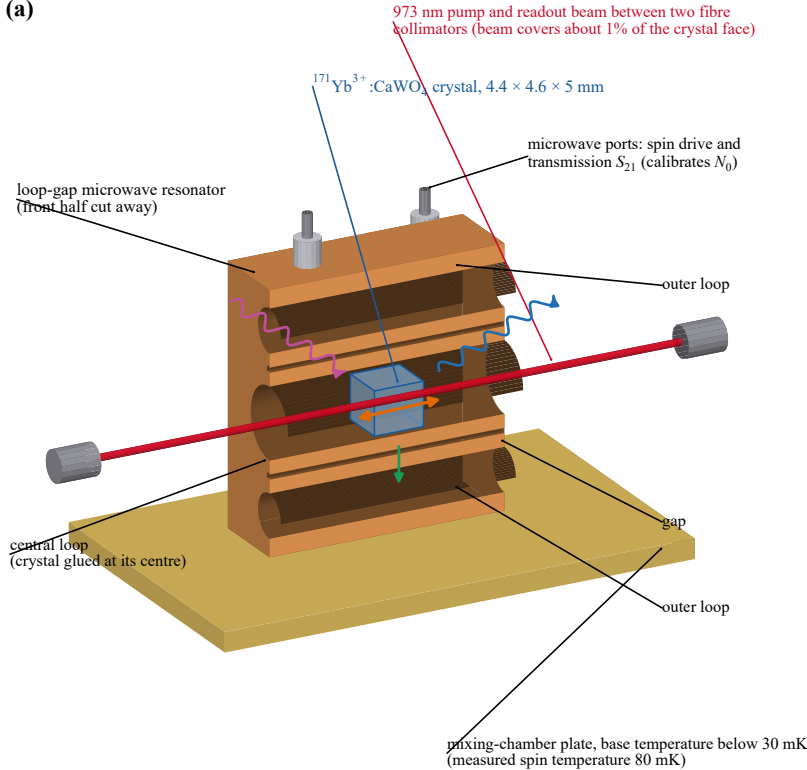


(a)**Loop-gap device (as built):**

- loop-gap resonator, $\kappa/2\pi = 660$ kHz
- $^{171}\text{Yb}^{3+}:\text{CaWO}_4$, 4.96 ppm ^{171}Yb , c -cut faces
- 973 nm light: pumping into $|\downarrow\rangle$, absorption readout
- microwave ports and fibre collimators
- mixing-chamber plate of a dilution refrigerator

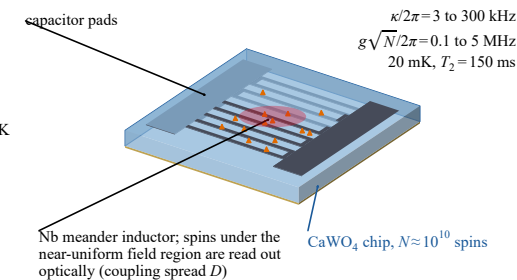
Parameters:

$\omega_s/2\pi = 3.084$ GHz, $\gamma_{\text{inh}}/2\pi < 5$ kHz, $T_2 > 150$ ms
 $g/2\pi = 15$ mHz, $V_m = 275$ mm³ (simulated), $N_0 \leq 7 \times 10^{14}$
 $\Delta/2\pi = 22$ MHz, $2\Delta/\kappa = 67$, $n_{\text{th}} = 0.19$ at 80 mK
 model values: $\gamma_{\text{inh}}/2\pi = 5$ kHz, $T_2 = 150$ ms

Processes in the model:

- cavity-mediated one-axis twisting, $\chi \hat{J}_+ \hat{J}_-$
- collective emission through the cavity, Γ_{SR}
- thermal cavity photons, rate $\Gamma_{\text{SR}} n_{\text{th}}$
- single-spin dephasing, $1/T_2$

(b) planar superconducting resonator on the same crystal: the design-map case (proposed, not built)

**(c)**